



University of Central Punjab

(Incorporated by Ordinance No. XXIV of 2002 promulgated by Government of the Punjab)

FACULTY OF INFORMATION TECHNOLOGY

3 – Input OR Gate Using Transistor

Project Description

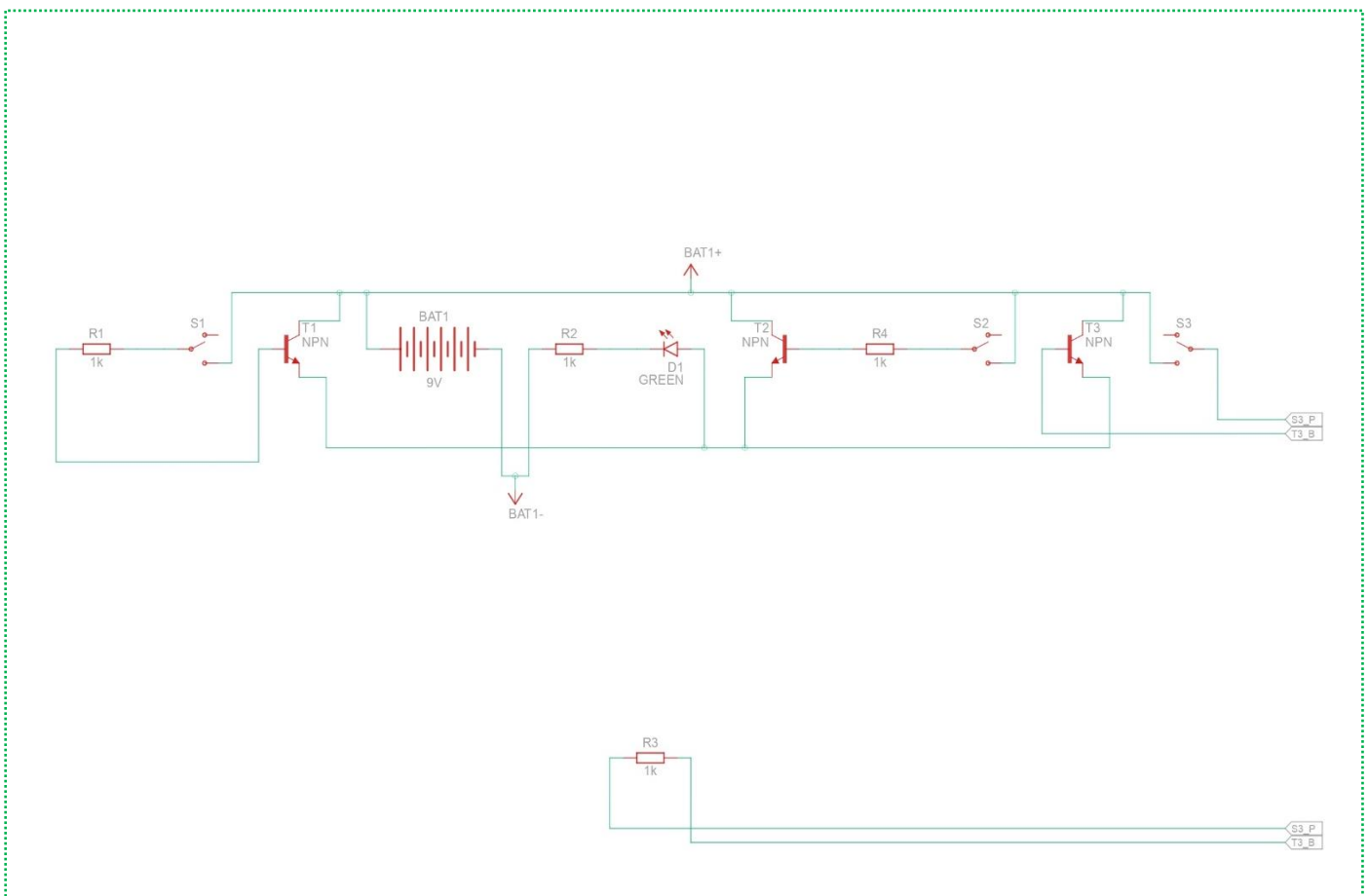
Project Name	3 – Input OR Gate Using Transistor
Course Title	Basic Electronics - LAB
Submission Date	15-July-2025
Faculty	Faculty Of Information Technology
Software Used	Tinkercad

1. Executive Summary :

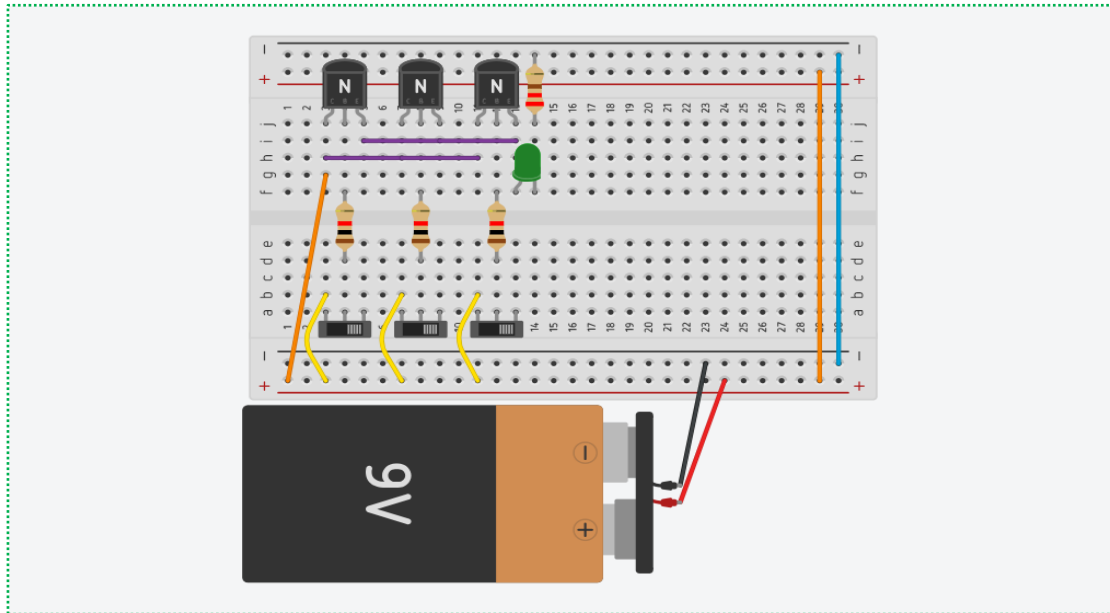
This project aimed to design and implement a **3-input OR gate using NPN transistors** on a breadboard. The objective was to understand how digital logic can be realized using basic electronic components instead of ICs. The expected outcome was a working OR gate where the **LED turns ON if any of the three inputs is HIGH (1)**.

- The project successfully replicated the OR gate functionality: the output was **HIGH** (LED ON) when **either or both inputs were HIGH**, and **LOW** (LED OFF) when **both inputs were LOW**.

2. Methodology - Circuit Diagram :



Circuit Diagram On Tinkercad :



Components Used :

- **Breadboard**
- **3 NPN Transistors (T1 & T2 & T3)**
- **3 Switches (S1 & S2 & S3)** as inputs
- **Resistors (R1 to R4)** to limit base and LED current
- **1 Green LED (D1)** for output visualization
- **9V DC** power source



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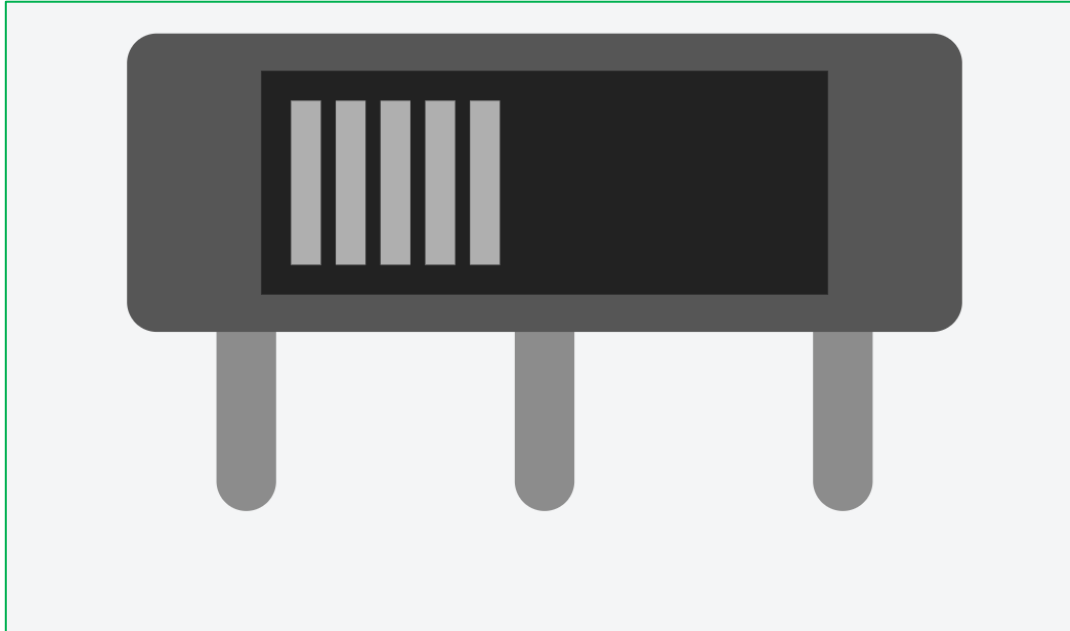


Figure 3: Sliding Switch

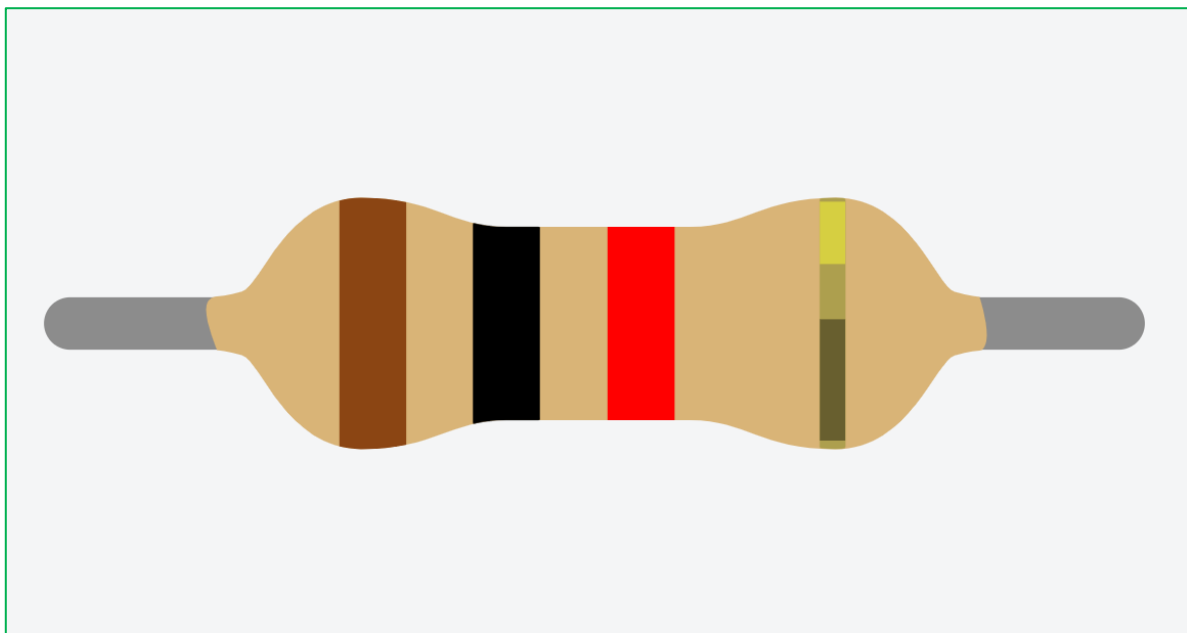


Figure 4: Resistor 1k Ohm

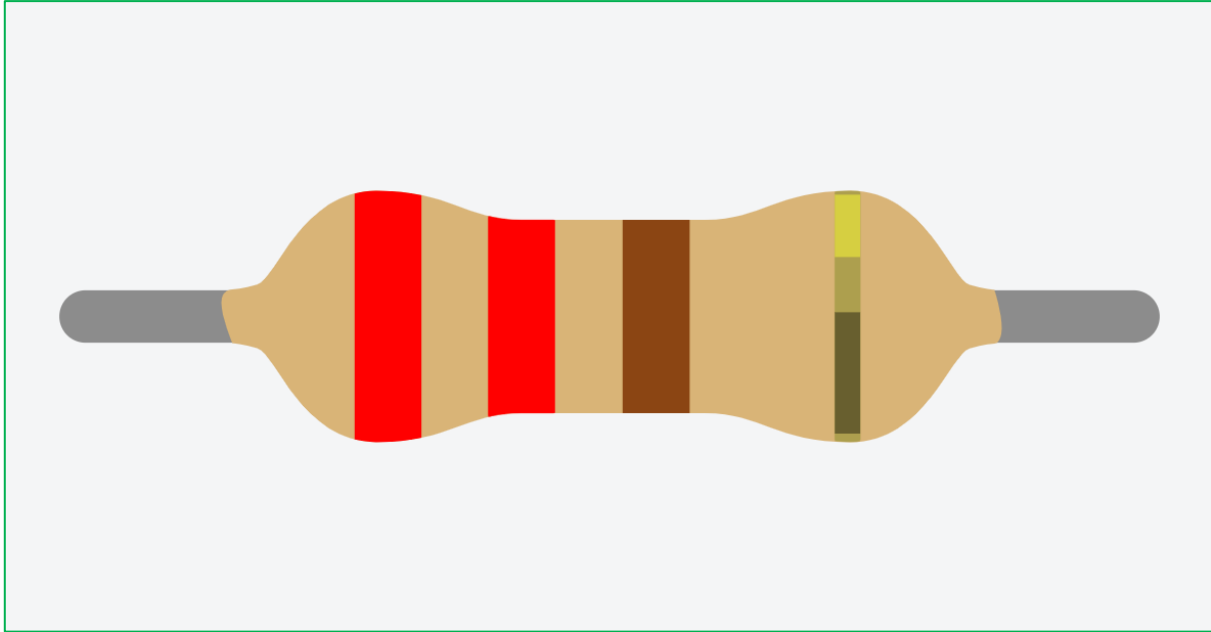


Figure 5: 220 Ohm Resistor



Figure 6: Green LED Bulb

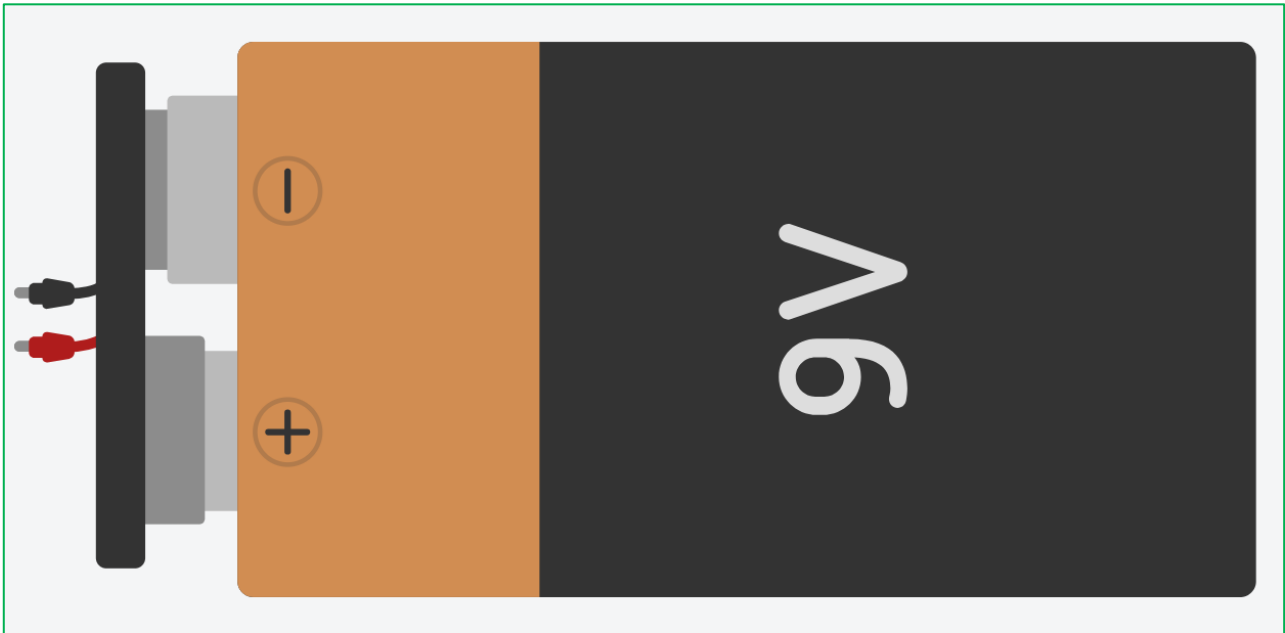


Figure 7: 9V Battery VSource

3. Implementation Steps :

1. Connected **V+ (9V)** and **V- (GND)** as the main power supply rails on the breadboard.
2. Placed **three NPN transistors (T1, T2, and T3)** in **parallel configuration** for output switching.
3. Connected the base of each transistor to individual **resistors (R1, R2, R3)**, followed by **slidebuttons S1, S2, and S3 (inputs A, B, C)**.



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4. The collectors of all three transistors were connected together and wired in series with a **green LED** and a **resistor**.
5. The emitters of all **three transistors** were connected directly to **GND**.
6. **Tested the output** by sliding each input switch and observing **LED** behavior.

4. **Results and Findings :**

- If no switch is slide to on → **LED OFF**
- If **S1** is slide to on → **LED ON**
- If **S2** is slide to on → **LED ON**
- If **S3** is slide to on → **LED ON**
- If **all three** are slided to on → **LED ON**

This confirms the **OR gate behavior** using basic components.



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Switch S1 (A)	Switch S2 (B)	Switch S3 (C)	LED Output (OR)
0 (OFF)	0 (OFF)	0 (OFF)	0 (OFF)
1 (ON)	0 (OFF)	0 (OFF)	1 (ON)
0 (OFF)	1 (ON)	0 (OFF)	1 (ON)
0 (OFF)	0 (OFF)	1 (ON)	1 (ON)
1 (ON)	1 (ON)	0 (OFF)	1 (ON)
0 (OFF)	1 (ON)	1 (ON)	1 (ON)
1 (ON)	0 (OFF)	1 (ON)	1 (ON)
1 (ON)	1 (ON)	1 (ON)	1 (ON)



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5. **Positive Outcomes :**

- ✓ Successfully built a **3-input OR gate** using basic discrete components.
- ✓ **Verified OR logic** functionality through physical testing.
- ✓ Gained practical knowledge of **transistor switching** behavior.
- ✓ Developed hands-on **understanding of logic gates** without using ICs.
- ✓ Improved **circuit design and troubleshooting** skills.